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Short-term effects of traditional Thai massage on electromyogram, muscle tension and pain among patients with upper back pain associated with myofascial trigger points

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ABSTRACT

Objective: To investigate effects of traditional Thai massage (TTM) on electromyographic (EMG) activity, muscle tension and pain intensity in patients with upper back pain associated with myofascial trigger points (MTrPs).
Design: A single-blind, randomized clinical trial.
Setting: The Department of Physical Therapy, School of Health Science, Mae Fah Luang University, Thailand.
Intervention: Fifty patients were randomly assigned to receive a 30-min session of either TTM or control (sham microwave diathermy).
Outcomes: Electromyogram (EMG), Muscle tension rating, and pain intensity rating
Results: TTM were associated with significant decreases in EMG, muscle tension and pain intensity after the end of treatment session ($p < 0.05$). For all outcomes, similar changes were not observed in the control group ($p > 0.05$) except for muscle tension ($p < 0.05$). In addition, there was a significantly greater reduction in all parameters for the TTM group when compared with the control group.
Conclusion: We therefore suggest that TTM can increase physical relaxation and reduce pain in patients with upper back pain associated with MTrPs.
Trial Registration: ClinicalTrials.gov Identifier: NCT02067325.
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1. Introduction

Myofascial pain syndrome (MPS) is a common musculoskeletal condition which is characterized by localized and referred pain, decreased range of motion, sleep problems and the presence of myofascial trigger points (MTrPs). A MTrP is a hyperirritable spot located in a palpable taut band of skeletal muscle that is painful on compression or stretch and produces a local twitch in response to snapping of the band.[1] The epidemiological study conducted in Thailand regarding musculoskeletal disorders found that MPS was highly prevalent and was the primary diagnosis in 36% of 431 patients.[2] A study by Skootsky et al.[3] found that MPS in the upper part of the body was more common than in other parts. Additionally, Rachlin and Rachlin[4] reported that most common sites of MPS were found in the upper trapezius, levator scapulae and axial postural muscles, respectively.

At present, the pathophysiology of MPS remains obscure making it difficult to choose the appropriate treatment for these patients, however, there are two theories; the motor endplate hypothesis and the energy crisis theory[5,6] might provide beneficial information for therapist in order to design an effective treatment. Simons[7] suggested that an effective intervention of MPS can be designed based on motor endplate hypothesis. He postulated that the MTrP pressure release (gentle thumb or finger pressure against the palpable tissue barrier in the MTrP) can normalize the length of shortened sarcomeres (also known as contraction knot) in the muscle fiber that causes release of MTrP.

Traditional Thai massage (TTM) is a popular branch of traditional Thai medicine.[8] It is a form of deep pressure massage with brief sustained compression on the targeted muscles along with spe-

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cific passive stretching.[9,10] The therapist uses his own body weight to apply thumb or digital pressure to the body of a patient along imaginary lines called "Sen Sib" and stretches the muscles at the end of massage session.[9] A previous study[10] found that most trigger points throughout the body fall along the "Sen Sib". Thus, it is believed that TTM could be an effective treatment to release MTrPs. TTM is usually conducted on a floor mat with the patient fully clothed.[9] Chaithavuthi and Muangsiri[11] claimed that TTM can improve the depth of breathing and relaxation, remove toxins from muscle mass, relieve muscle tension, relax tendons and enhance elasticity of muscle. Moreover, Eungpinichpong[12] reported that TTM can improve physical and mental relaxation, improve blood circulation, decrease pain and muscle tightness, and increase range of motion and muscle flexibility in several musculoskeletal disorders. Although the indigenous use of TTM for relieving various conditions has been practiced in Thailand for a long period of time, strong evidence to support its beneficial effects is still not enough. Thus, TTM needs more reliable scientific evidence to confirm its outcomes, especially its relevant physiological effects.

Nowadays, there are some research studies that have investigated the physiological effects of TTM on different conditions.[13–15] However, the research to support the effect of TTM on electromyographic (EMG) activity, which can provide direct and reliable information regarding the physical relaxation of the muscle, is lacking. The current study evaluated the immediate effects of TTM on EMG activity, feeling of muscle tension and pain intensity in patients with upper back pain associated with MPS.

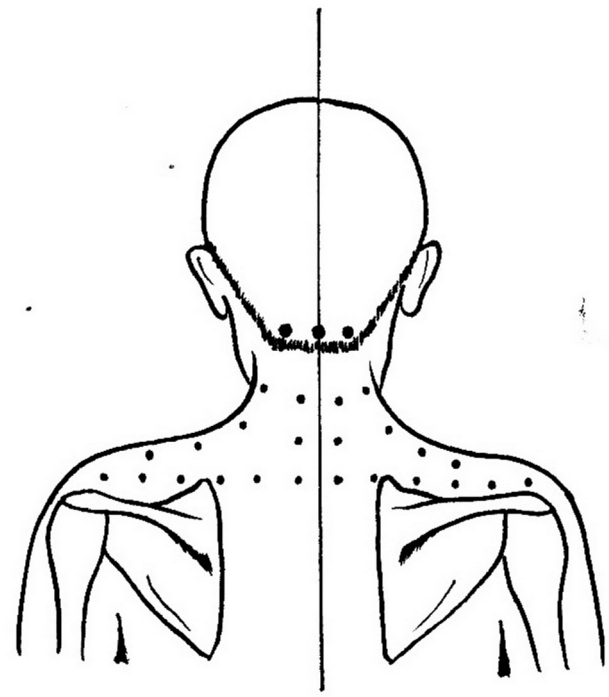

Fig. 1. The massage points are located along three lines on each side of the body and at an additional, three points along the superior nuchal line.

2. Methods

2.1. Design

A prospective, parallel group, assessor-blind, randomized controlled trial was conducted in the Department of Physical Therapy, School of Health Sciences, Mae Fah Luang University, Chiang Rai Province, Thailand. Patients with upper back pain associated with MTrPs were randomly allocated to either a TTM or a control group (sham microwave diathermy: MWD) using the permuted-block randomization in blocks of 2, 4 and 6 subjects (the STATA software). A list was generated by one of the researchers and used by a research assistant. Patient assignment to a group was determined through sealed envelope distribution and investigators were blinded. Outcomes were assessed baseline and immediately after the treatment sessions. The primary outcome measure was the EMG activity and the secondary outcomes included feeling of muscle tension and pain intensity. The study was approved by the ethics committee of Mae Fah Luang University. Written informed consent was obtained from all patients.

2.2. Participants

Patients with upper back pain associated with MTrPs were recruited from Chiang Rai province, Thailand during a 4-month period between February 2014 and May 2014 through advertisements on bulletin boards. Inclusion criteria were patients aged between 18 and 50 years with spontaneous upper back pain for longer than 3 months, and with at least one MTrP present in the upper trapezius muscles. The criteria used to diagnose MTrP described previously by Buttagat et al.[16] was as follows; the presence of a tender nodule that causes a referred pain within taut bands of muscle in areas which the patient identified as painful.

The present study excluded patients with a history of cervical radiculopathy, a history of neurological disorders and a diagnosis of conditions that could be contraindicated for TTM, such as fever, contagious skin disease, injury or inflammation of muscle, bone fracture and/or joint dislocation, open wound, uncontrolled hypertension, Thrombophlebitis, Hemophilia, drug and/or alcohol intoxication.

Sample size determination was conducted using an analysis of covariance (ANCOVA) formula.[17] The calculation was based on our pilot study that compared the effect of TTM (n = 10) with that of sham MWD (Control group) (n = 10) for patients with upper back pain associated with MTrPs. A standard deviation (of EMG activity) of 0.0027 mV for the TTM group and of 0.0064 mV for the sham MWD group was observed. These variances were used to calculate the sample size needed to detect a 0.0030 mV change in EMG activity with 90% power and 5% significance. According to these criteria, 50 patients were needed.

2.3. Interventions

The patients in the TTM group received a 30 min session of TTM onto the upper back area while lying on their side.

All TTM in the current study was performed by a well-trained massage therapist. Massage points included in this method are located along three lines on each side of the body and at an additional, three points along the superior nuchal line (Fig. 1). The pressing technique employed in TTM used the body weight of the massage therapist to apply gentle and gradually increasing pressure through the thumbs or fingers for 5–10 s per point. The amount of pressure did not exceed the pain pressure threshold of each patient. This sequence could be repeated several times for each massage point. Specific stretching technique for the upper back pain patients was applied passively at the end of the session to conclude the treatment (Fig. 2). The TTM technique and procedure for upper back pain patients used in the current study has been described in more detail by Eungpinichpong.[12]

The patients in the control group received one 30-min session of sham MWD (in the same environment as the TTM group) using the Enraf-Nonius Radarmed 950 Plus (Delft, The Netherlands) with a large field radiator placed 15 cm from the upper back while a patient lay on their side. The MWD device was turned on but kept

Fig. 2. Stretching techniques of traditional Thai massage for patients with upper back pain associated with myofascial trigger points.

in standby mode during 30 min. Furthermore, at the end of the study, all patients in the control group were given the opportunity for instruction in a series of upper back and scapular exercises to conduct at home.

The patients in both groups did not receive any concomitant treatment. Each of the groups was treated in separate rooms in order to decrease any confounding factors.

2.4. Outcome measures

All outcome measures were assessed and analyzed by one of the research assistants who was blinded and was not obligated to provide any treatment for patients.

The EMG activity of upper trapezius muscle was measured using the BIOPAC system (*BIOPAC* Systems, Inc., Goleta, California). The EMG signals were analyzed using Acknowledge Software version 4.2 (*BIOPAC* Systems, Inc., Goleta, California). Patient rested in a prone position at room temperature (25 °C) for 10 min before EMG recording in order to ensure that a resting state was achieved. The electrode placement used in the present study has been described previously in detail.[18–20] The skin was prepared by shaving and cleansing the area with 70% isopropyl alcohol. The EMG RMS was collected before and immediately after the treatment period. Analysis was based on a 5-min period of EMG signal acquisition. Data were notch filtered at 50 Hz, high pass filtered at 5 Hz and low pass filtered at 500 Hz. The following parameters were used: band width 5–500 Hz, input impedance 5 MΩ, sampling rate 1000 Hz and gain 1000. EMG data were not normalized because all comparisons made in the present study were within-day comparisons of single muscles.[21–23]

Pain intensity and feeling of muscle tension were measured using separate 10-cm visual analogue scale (VAS). The scale ranged from 0 to 10, where 0 indicated "no pain", or "no muscle tension", respectively, and 10 indicated "the most pain ever" and "the tensest ever experienced", respectively. Patients were requested to mark the scale indicating their level of pain intensity or muscle tension.

2.5. Data analysis

Statistical analysis was conducted using STATA Version 10 software (StataCorp LP, College Station, Texas). Mean and standard deviations or 95% confidence intervals of the values were calculated for each variable. Paired *t*-tests were used to compare the outcome variables immediately before and after the treatment period within each respective group. An analysis of covariance (ANCOVA) was used to compare the difference in post-test values between groups after adjusting for differences in baseline values and to estimate the adjusted post-test mean value for each outcome measure of each group. *P*-values less than 0.05 were considered as statistically significant.

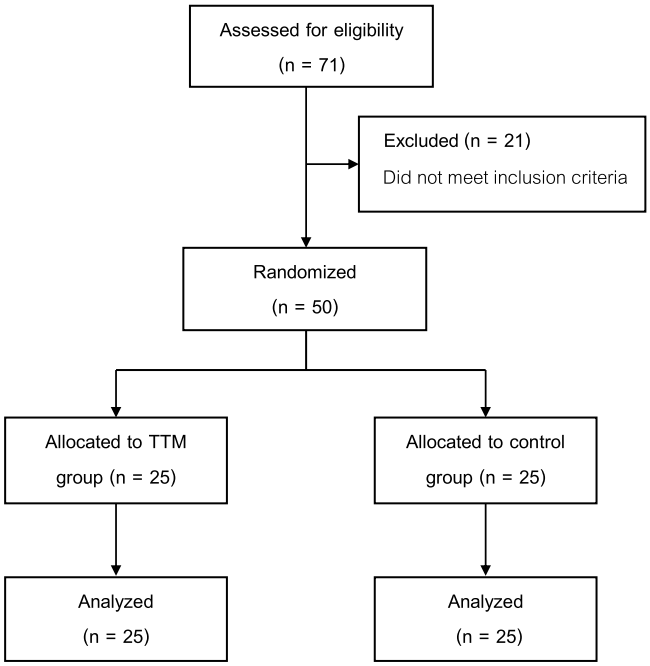

Fig. 3. Participants flowchart.

3. Results

Participant flow is shown in Fig. 3. The study screened 71 patients. Fifty of these patients met the inclusion criteria and were recruited (25 patients were randomized to TTM and 25 patients to sham MWD). The baseline data of demographic characteristics and baseline clinical characteristics of the patients are summarized in Table 1. Baseline characteristics of the study sample were equally balanced between groups.

3.1. Immediate effects of TTM vs. Sham MWD

Table 2 shows that immediately after receiving TTM, the EMG activity, feeling of muscle tension and pain intensity significantly decreased compared with baseline values (<0.001). On the other hand, no statistically significant difference was found for the sham MWD (Control group) except for feeling of muscle tension (Table 2). A comparison of the adjusted post-test values for all outcome measures between groups indicated a significantly greater improvement in EMG activity, feeling of muscle tension and pain intensity for the TTM group compared with the control group ($p < 0.05$) (Table 3).

4. Discussion

To the best of our knowledge, this is the first randomized, single-blind, clinical trial examining the effects of TTM on EMG activity in patients with upper back pain associated with MTrPs and is the first research that evaluates the physiological changes of the muscle directly after receiving TTM. In order to confirm the effect of

Table 1
Demographic and baseline clinical characteristics of patients with upper back pain associated with MTrPs.

Variable	Control group (n = 25)	TTM group (n = 25)
Age (years); Mean (SD) (Range)	22.76 (4.10) (18–35)	21.72 (2.05) (19–30)
Gender (number of females)	22	21
Weight (kg); Mean (SD)	58.24 (13.40)	59.32 (15.99)
Height (centimeters); Mean (SD)	161.32 (6.73)	160.48 (6.49)
Body Mass Index (kg/m^2); Mean (SD)	22.23 (4.04)	22.92 (5.53)
Severity of upper back pain: by pain scale; mean (SD)	4.50 (2.65)	4.97 (2.00)
Duration of upper back pain episode (months); Mean (SD) (Range)	22 (16.5) (3–53)	20 (17.3) (3–59)

Table 2
Comparison of outcome measures between baseline and post-test assessments in the TTM and control groups (Dependent *t*-tests).

Parameter	Control group				TTM group			
	Baseline (Mean (SD))	Post–test (Mean (SD))	Difference (95% CI)	P value	Baseline (Mean (SD))	Post–test (Mean (SD))	Difference (95% CI)	P value
EMG (RMS):mV	0.0078 (0.0062)	0.0086 (0.0076)	−0.0008 (-0.0019– 0.0002)	0.12	0.0143 (0.0139)	0.0114 (0.0122)	0.0029 (0.0016–0.0042)	<0.001
Muscle tension (VAS):cm	5.38 (2.59)	4.30 (2.93)	1.07 (0.26–1.89)	0.01	5.32 (2.04)	3.19 (1.96)	2.13 (1.33–2.92)	<0.001
Pain intensity (VAS):cm	4.50 (2.65)	3.84 (2.65)	0.65 (-0.85 –1.39)	0.08	4.97 (2.00)	2.77 (1.94)	2.21 (1.56–2.86)	<0.001

Table 3
Comparison of mean post-test measures between TTM and control groups after adjustment for differences in baseline values (ANCOVA).

Parameter	Control group	TTM group	Difference (95% CI)	P value
EMG (RMS):mV	0.0116	0.0084	0.0032 (0.0016–0.0048)	<0.001
Muscle tension (VAS):cm	4.28	3.21	1.07 (0.003–2.13)	0.049
Pain intensity (VAS):cm	4.02	2.59	1.43 (0.52–2.34)	0.003

TTM on muscle activity, the feeling of muscle tension and pain intensity were also assessed. The results of this study suggest positive effects of TTM on patients with back pain associated with MTrPs, as indicated by a decrease in EMG activity, feeling of muscle tension and pain intensity (refer to Tables 2 and 3). According to these findings, TTM provides both objective and subjective benefits for MPS patients and the outcomes of this study strongly support the anecdotal evidence regarding the use of TTM for reducing muscle tension and muscle stiffness and improving physical relaxation.[11,24]

The finding of the present study also revealed that patients receiving TTM had significantly lower EMG activity, muscle tension and pain intensity than those in the control group. Thus, we can suggest that treatment by TTM in patients with upper back pain associated with MPS is superior to the sham MWD.

The results of the present study are generally in line with those of previous studies. Buttagat et al.[13] reported that after the MPS patients received TTM, pain intensity, pressure pain threshold (PPT), muscle tension and body flexibility improved. Chatchawan et al.[25] found that TTM decreased pain intensity and increased PPT, back performance and body flexibility in patients with back pain associated with MTrPs. Mackawan et al.[15] revealed that a 10 min session of TTM can decrease pain intensity and substance P level among patients with nonspecific low back pain. In addition, Aguilera et al.[26] showed that the pressure point compression decreased EMG activity and MTrPs sensitivity and increased cervical ROM in patients with trapezius MTrPs. Anywise, it should be noticed that some aspects of these studies, including the characteristics of participants, the duration of massage, the massage technique, the body area treated and the position of patients during treatment, differ from this study.

The possible mechanism of TTM that may lead to a decrease in EMG activity and feeling of muscle tension might be explained by the findings of Sefton et al.[27] and Oksa et al.[28] They suggested that the massage can decrease alpha-motoneuron pool excitability which might lead to decreased muscle electrical activity and muscle tension. Another potential mechanism by which TTM promotes physical relaxation and decreases muscle tension is that the massage can result in lengthening of the sarcomeres of skeletal muscle leading to a breaking apart of stable actin/myosin cross bridges resulting in a decrease of muscle tension.[9,29]

For the reduction of pain intensity resulting from the TTM, several theories could be proposed to describe the feasible mechanisms underlying the effects of TTM. Mackawan et al.[15] postulated that the reduction of pain by TTM can be explained by gate control theory. Eungpinichpong and Kongnaka[30] reported that TTM can increase blood flow to the muscles to be treated and surrounding area, consequently removing pain metabolites e.g. substance P and decreasing pain.[15] Moreover, boosting the serotonin (pain-relieving neurotransmitters) levels resulting from TTM might be another mechanism.[14,31,32]

Potential limitations associated with the current study include the lack of blinding of patients to the type of treatment being received. However, such blinding could not easily be performed due to the nature of non-pharmacological treatments. Another limitation was that the present study did not look at the long-term effects of TTM. The evaluation of long-term effects is difficult to conduct because the EMG activity is quite sensitive. Changing of environments, physical activities or exercise behaviors of each patient during the waiting period for a follow-up measurement could result in unreliable data which can lead to wrong research results. Thus, the long-term effects evaluation with strict control of environments, physical activities and exercise behaviors before a follow-up measurement would be an interesting direction for future study. In addition, most patients who participated in the present study were young. Therefore, the results of the current study may not translate to patients of other age groups. Future study should involve participants of different ages.

4.1. Clinical implication

The result of the present study suggests that TTM has immediate benefits on EMG activity, muscle tension and pain among patients with upper back pain associated with MTrPs. Therefore, TTM might

be considered to be a useful intervention for promoting physical relaxation and decreasing muscle tension and pain in MPS patients.

5. Conclusion

The finding of the present study has indicated that a single treatment of TTM onto the upper back region has shown an immediate significant decrease in EMG activity, feeling of muscle tension, and pain intensity in patients with upper back pain associated with MTrPs. Since TTM is safe and easy for therapists to teach patients or family caregivers to use, we suggest that TTM should be considered as an interesting intervention option for promoting physical relaxation and decreasing muscle tension and pain in this patient population.

Conflict of interest statement

No conflict of interest is declared.

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